

SAT UZBEKISTAN

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Module-1

1.

A jogger starts running from a point A with coordinates $(0, 0)$ and reaches point B with coordinates $(3, 9)$. He turns 90 degrees to the right relative to his initial direction and continues running in a straight line at the same distance as he ran between point A and B , reaching and stopping at point C . What is the final coordinate of point C ?

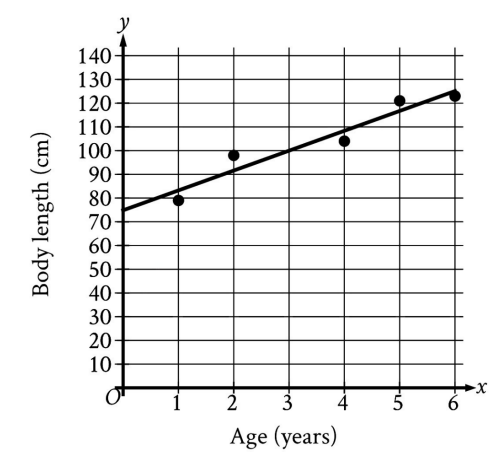
A: $(2, 10)$

B: $(-9, 3)$

C: $(12, 6)$

D: $(-2, 10)$

2.



The scatterplot shows 5 measurements of the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 year to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length predicted by the line of best fit, to the nearest 10 cm?

3.

There is a linear relationship between the mass of an object attached to a vertical spring and the length of the spring extension. The table shows data collected after certain objects were each attached to the spring.

Mass of object	Length of spring extension
20 grams	7 centimeters
60 grams	21 centimeters

What is the length of the spring extension, in centimeters, when a 50-gram object is attached to the spring?

4.

A right square pyramid has a height of 6 units and a volume of 128 cubic units. What is the length, in units, of one side of the base of the pyramid?

5.

$$8x + 32y = 30$$

$$12x + 48y = 45$$

ER

At how many points do the graphs of the given equations intersect in the xy -plane?

- A) Zero
- B) Exactly one
- C) Exactly Two
- D) Infinitely many

6.

In the xy -plane, the graph of the equation $(x - 4)^2 + (y - 9)^2 = 16$ is a circle. The point $(8, c)$, where c is a constant, lies on this circle. What is the value of c ?

7.

$$y > \frac{9}{7}x + b$$

In the given inequality, b is a positive constant. Which of the following does NOT contain any points (x, y) in the xy -plane that are solutions to this inequality?

☐ A) The region where $x < 0$ and $y < 0$

☐ B) The region where $x < 0$ and $y > 0$

☐ C) The region where $x > 0$ and $y > 0$

☐ D) The region where $x > 0$ and $y < 0$

8.

The expression $x^2 + kx + 55$, where k is a constant, can be rewritten as $(x + n)(x + 11)$, where n is a constant. What is the value of k ?

9.

A beaker containing a liquid is placed on a table. The function $g(t) = 297 + (363 - 297)(2.72)^{-0.103t}$ gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

10.

The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy -plane, the graph of $y = f(x)$ has a y -intercept at $(0, -22)$ and passes through the point $(2, 26)$. What is the value of $a + b$?

11.

$34x + 79y = 260$

The given equation represents the volume of mulch x , in cubic feet, and the volume of soil y , in cubic feet, in a mixture of mulch and soil that weighs 260 pounds. If the volume of soil in the mixture is 2 cubic feet, what is the volume of mulch in the mixture, in cubic feet?

12.

$y < x$

$y > -4x - 9$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
5	4
6	5
7	6

B)

x	y
-12	-5
-11	-6
-10	-7

C)

x	y
-12	-13
-11	-12
-10	-11

D)

x	y
5	10
6	9
7	8

13.

$$f(x) = 5(g(x)) - 3$$

$$g(x) = |13x - 9|$$

ER

The functions f and g are defined by the given equations. What is the value of $f(-10)$?

14.

$$\sqrt{x^2 - 95x + 300} = x\sqrt{11}$$

What are all possible solutions to the given equation?

15.

The population of a certain city doubled every 75 years from 1657 to 1957. The population of this city was 240,000 in 1957. What was the population of this city in 1657?

16.

A partially filled container containing 26 milliliters of water is placed under a leaky faucet that produces one 0.03-milliliter drop of water every 3 seconds. Until the container is full, which of the following can be used to represent the volume v , in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 3?

A) $v = 0.01t + 26$

B) $v = 0.09t + 26$

C) $v = 3t$

D) $v = 0.03t + 26$

17.

At the beginning of an experiment, the temperature of a liquid is 23°C . During the first 4.0 minutes of the experiment, the temperature of the liquid increases at an average rate of 6.5°C per minute. Then, the temperature of the liquid increases at a constant rate of 2.2°C per minute. If the temperature of the liquid reaches 61°C x minutes after the beginning of the experiment, where $x > 4.0$, which equation represents this situation?

- A) $2.2x + 49 = 61$
- B) $2.2(x - 4) + 49 = 61$
- C) $2.2(x - 4) + 26 = 61$
- D) $2.2x + 23 = 61$

18.

A science club from a certain high school in Ohio conducted an experiment to study the effects of smell on taste. A sample of 100 students was selected at random from the high school to participate in the experiment. First, the students tasted a pureed food sample while wearing covered goggles and nose plugs. Then, the students tasted the same pureed food sample without nose plugs. 35 students were able to correctly identify the pureed food sample while wearing nose plugs. Which of the following is the largest population to which the results of the experiment can be generalized?

- A) All the students from the high school
- B) The 100 students who participated in the experiment
- C) The 35 students who were able to correctly identify the pureed food sample while wearing nose plugs
- D) All students from high schools in Ohio

19.

For the positive quantities h , j , and k , 13% of h is equivalent to 19% of j , and j is equivalent to 91% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

20.

The function f is defined by $f(x) = 11x^3$. The graph of $y = f(-x) + c$ in the xy -plane, where c is a positive integer constant, has an x -intercept at $(r, 0)$ and a y -intercept at $(0, t)$, where r and t are constants. Which of the following must be true about r and t ?

☐ A) $r < 0$ and $t < 0$

☐ B) $r < 0$ and $t > 0$

C) $r > 0$ and $t > 0$

☐ D) $r > 0$ and $t < 0$

21.

A certain investment account offers a special interest rate for the first 6 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$600 and did not make any other deposits or withdrawals. 6 months after Bennett opened the account, the balance had increased by 0.4% of the original balance. 8 months after Bennett opened the account, the balance had increased by an additional 0.3% of the balance at the end of the first 6 months. Every 2 months after the first 8 months, the balance had increased by an additional 0.3% of the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 6$?

☐ A) $B(x) = 602.40(1.003)^{\frac{x}{2}-6}$

$B(x) = 602.4(1.003)^{\frac{(x-6)}{2}}$

☐ C) $B(x) = 602.40(1.003)^{2x-12}$

☐ D) $B(x) = 602.40(1.003)^{2x-6}$

22.

$$\frac{1}{24}x^2 + \left(s - \frac{1}{24}t\right)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k ?

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Module-2

1.

In triangle RST , angle T is a right angle, point L lies on RS , point K lies on ST , and LK is parallel to RT . If the length of RT is 72 units, the length of LK is 24 units, and the area of triangle RST is 396 square units, what is the length of KT , in units?

2.

The area of a rectangular region is increasing at a rate of 210 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

- A) 0.33
- B) 1.07
- C) 11.48
- D) 19.52

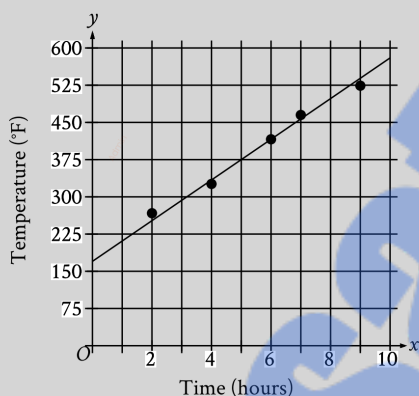
3.

If $\frac{x+5}{3} = \frac{x+5}{13}$, the value of $x + 5$ is between which of the following pairs of values?

- A) 7 and 13
- B) 4 and 16
- C) -4 and 4
- D) -6 and -3

4.

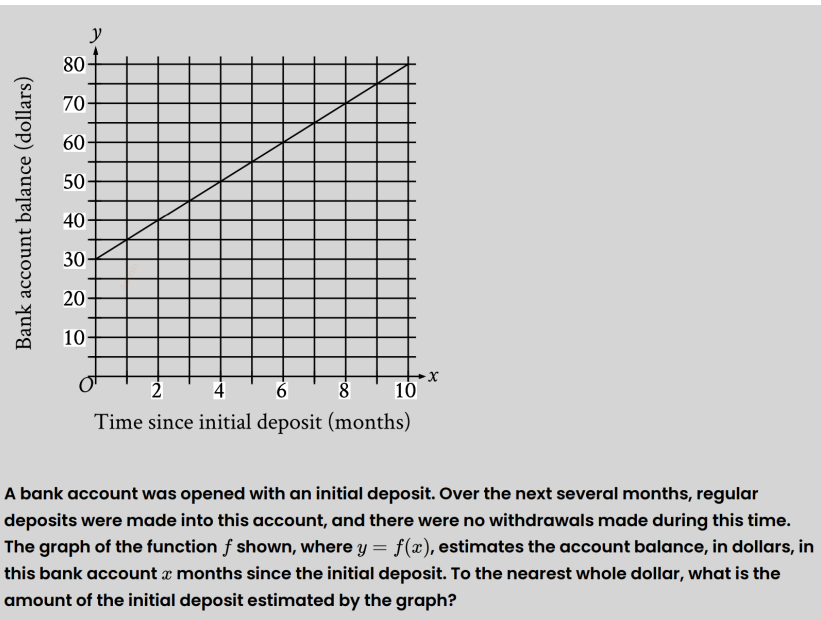
For the first 10 hours of an experiment, the scatterplot shows the temperature y , in degrees Fahrenheit ($^{\circ}\text{F}$), of an object at various times x , in hours, since the start of the experiment. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit?

- ☐ A) The predicted temperature increases at a constant rate of approximately 170°F per hour over the 10-hour period.
- ☐ B) The predicted temperature decreases at a constant rate of approximately 170°F per hour over the 10-hour period.
- ☐ C) The predicted temperature increases at a constant rate approximately 40 F per hour over the 10-hour period
- ☐ D) The predicted temperature decreases at a constant rate of approximately 40°F per hour over the 10-hour period.

5.



6.

$$f(x) = -16x^2 + 76$$

The function f gives the estimated height, in feet, of an acorn x seconds after the acorn fell from a tree. Based on the function, what is the estimated height, in feet, of the acorn before it fell from the tree?

7.

During hibernation, American black bears do not eat or replenish calories. A certain black bear weighed 515 pounds when entering hibernation and lost weight at a rate of 0.5 pound per day. At this rate, how many days after entering hibernation did the black bear weigh 490 pounds?

8.

A freight elevator can hold a maximum weight of 5,450 pounds during one trip. A 190-pound person needs to deliver several boxes using the freight elevator. Some of these boxes weigh 23 pounds each and the others weigh 60 pounds each. Which inequality represents the possible combinations of the number of 23-pound boxes, x , and the number of 60-pound boxes, y , the person can deliver during one trip if only the person and the boxes are on the freight elevator?

- ☐ A) $60x + 23y \geq 5,450$
- ☐ B) $23x + 60y \geq 5,260$
- ☐ C) $60x + 23y \leq 5,450$

☐ D) $23x + 60y \leq 5,260$

9.

The density of a certain type of marble stone is 2.6000 grams per cubic centimeter. If a sample of this type of stone is in the shape of a sphere with a diameter of 31.000 centimeters, what is the mass of this sample, in grams, to the nearest whole number? (Use 3.14159 for π .)

10.

The function w is defined by $w(r) = \frac{1}{r-8} - \frac{r-5}{-r+4.25}$. What is the greatest possible value of r such that $w(r) = 0$?

11.

$$1.5c + 3d = 108$$

The given equation describes the relationship between the number of cats, c , and the number of dogs, d , that can be cared for at a pet care business during a week. If the business cares for 36 dogs during a week, how many cats can it care for during this week?

12.

Stefan purchased 500 feet of fencing. He used 80% of this fencing to surround a vegetable garden. How many feet of fencing did Stefan use to surround the vegetable garden?

13.

On January 1, 2000, the population of a town was 26,255 and on January 1, 2010, the population was 26,956. The equation $10x + 26,255 = 26,956$ describes this situation. Which of the following is the best interpretation of x in this context?

☐ A)

The total increase in population between 2000 and 2010

B) The average increase per year of the population between 2000 and 2010

☐ C)

The percentage by which the population of the town increased each year between 2000 and 2010

☐ D)

The projected population of the town 10 years after 2010

14.

Cube B has a volume of 125 cubic inches. The length of each edge of cube A is k times the length of each edge of cube B. If cube A has a volume of 3,375 cubic inches, what is the value of k ?

15.

A right square pyramid has a height of 6 units and a volume of 128 cubic units. What is the length, in units, of one side of the base of the pyramid?

16.

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 27% every 14.07 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

☐ A) $M = 100(0.27)^{\frac{t}{14.07}}$

☐ B) $M = 100(0.27)^{t+14.07}$

☐ C) $M = 100(0.73)^{\frac{t}{14.07}}$

☐ D) $M = 100(0.73)^{t+14.07}$

17.

A department store sells pants in three lengths: short, regular, and long. The table summarizes the distribution of pant length, by color, for all the pants in the store.

	Black	Brown	Blue	Total
Short	25	14	24	63
Regular	23	15	22	60
Long	20	19	19	58
Total	68	48	65	181

A pair of pants is selected at random. What is the probability of selecting a pair of pants that is brown, given that the pair of pants is regular length? (Express your answer as a decimal or fraction, not as a percent.)

18.

An equation of the line in the xy -plane that contains the points $(0, -8)$ and $(-2, 0)$ is $Ax + By = 9$, where A and B are constants. What is the value of B ?

19.

$$g(x) = \frac{1}{11}(6(2)^x + 7)$$

The function g is defined by the given equation. For all values of x , the value of $g(x)$ is greater than k , where k is a constant. What is the greatest possible value of k ?

20.

$$nx^2 - 16x = 26x^2 - 8$$

In the given equation, n is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of n ?

21.

A quadratic function gives the estimated length of daylight $d(t)$, in hours, in a certain city t months after March 1, where $0 \leq t \leq 7$. According to the function, the estimated length of daylight is 12.66 hours 6 months after March 1 and the maximum estimated length of daylight is 13.91 hours 3.5 months after March 1. Based on this function, what is the estimated length of daylight, in hours, on March 1?

22.

In triangle ABC , the measure of angle A is 52° and $AC = 40$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

☐ A) $AB = 45$ and $PQ = 45$.

☐ B) $AB = 45$ and $QR = 135$.

☐ C) The measures of angle B and angle R are 40 and 88 respectively

☐ D) The measures of angle B and angle Q are 52° and 40° , respectively.